LEC - 49

Earners AI and ML by Aditya Jain Sir - Numpy-1

Summary Notes

```
marks = [55, 90, 40, 70, 80]
bonus = 5

new_marks = []

for m in marks:
  new_marks.append(m+bonus)

print(new_marks)
```

```
[60, 95, 45, 75, 85]
```

```
marks + 5
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_2289/2290572357.py in <cell line: 0>()
----> 1 marks + 5

TypeError: can only concatenate list (not "int") to list
```

Next steps: Explain error

```
# NumPy --> Numerical Python

import numpy as np
bonus = 10
marks = np.array([55, 90, 40, 70, 80])

new_marks = marks + bonus

print(new_marks)
```

```
[ 65 100  50  80  90]
```

```
a = [4,6,8,2,1]

a[0]
```

```
4
```

```
a[1]
```

```
6
```

```
a.append(10)
```

```
a
```

```
[4, 6, 8, 2, 1, 10]
```

```
a = [4,6,8,2,1]
type(a)
```

```
list
```

```
print(a)
```

```
[4, 6, 8, 2, 1]
```

```
np_a = np.array(a)
print(np_a)
```

```
[4 6 8 2 1]
```

```
type(np_a)
```

```
numpy.ndarray
```

```
# given a python list:
a = [3,6,1,9]

b = []

for i in a:
  b.append(i*2)

print(b)
```

```
[6, 12, 2, 18]
```

```
type(b)
```

```
list
```

```
a_np = np.array(a)
b_np = a_np*2
print(b_np)
```

```
[ 6 12  2 18]
```

```
type(b_np)
```

```
numpy.ndarray
```

```
a_np-2
```

```
array([ 1,  4, -1,  7])
```

```
a_np+2
```

```
array([ 5,  8,  3, 11])
```

```
a_np/2
```

```
array([1.5, 3. , 0.5, 4.5])
```

```
a = [6, 12, 2, 18]
b = np.array(a)
```

```
a = [[2,4,5],
     [6,3,1],
     [5,7,2]]
type(a)
```

```
list
```

```
print(a)
```

```
[[2, 4, 5], [6, 3, 1], [5, 7, 2]]
```

```
b = np.array(a)
print(b)
```

```
[[2 4 5]
 [6 3 1]
 [5 7 2]]
```

```
type(b)
```

```
numpy.ndarray
```

```
a = np.zeros(5)
print(a)
```

```
[0. 0. 0. 0. 0.]
```

```
a = np.zeros((4,3))
print(a)
```

```
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

```
a = np.ones(4)
print(a)
```

```
[1. 1. 1. 1.]
```

```
a = np.ones((4,5))
print(a)
```

```
[[1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]
 [1. 1. 1. 1. 1.]]
```

```
a = np.arange(5)
a
```

```
array([0, 1, 2, 3, 4])
```

```
a = np.arange(1,11)
a
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10])
```

```
type(a)
```

```
numpy.ndarray
```

```
np.arange(1,11,2)
```

```
array([1, 3, 5, 7, 9])
```

```
np.arange(2,11,2)
```

```
array([ 2,  4,  6,  8, 10])
```

```
np.arange(5,11)
```

```
array([ 5,  6,  7,  8,  9, 10])
```

```
np.arange(1,32)
```

```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17,
       18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31])
```

```
np.linspace(0,10,5)
```

```
array([ 0. ,  2.5,  5. ,  7.5, 10. ])
```

Arithmetic Operations

```
a = np.array([3,5,6])
b = np.array([5,6,1])
```

```
print(a)
```

```
[3 5 6]
```

```
print(b)
```

```
[5 6 1]
```

```
a = [3,5,6]
b = [5,6,1]

# c = [8,11,7]
c= []

n = len(a)
for i in range(n):
  c.append(a[i] + b[i])

print(c)
```

```
[8, 11, 7]
```

```
a+b
```

```
[3, 5, 6, 5, 6, 1]
```

```
an = np.array(a)
bn = np.array(b)

print(an+bn)
```

```
[ 8 11  7]
```

```
print(an-bn)
```

```
[-2 -1  5]
```

```
print(a*b)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_2289/2818731505.py in <cell line: 0>()
----> 1 print(a*b)

TypeError: can't multiply sequence by non-int of type 'list'
```

Next steps: Explain error

```
print(an*bn)
```

```
[15 30  6]
```

```
print(an)
```

```
[3 5 6]
```

```
print(an**2)
```

```
[ 9 25 36]
```

```
print(an+5)
```

```
[ 8 10 11]
```

```
print(an-5)
```

```
[-2  0  1]
```

```
print(an*5)
```

```
[15 25 30]
```

```
print(an)
```

```
[3 5 6]
```

```
Start coding or generate with AI.
```